

Aditya Choudhury *M.S. Computer Science*

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PROFILE

Graduate computer science student specializing in backend systems, applied machine learning, and full-stack AI product development. Experienced building production ed-tech platforms with LLM-integrated features, adaptive learning models, and secure, scalable backend architecture serving hundreds of active users. Strong analytical thinker with proven ability to take AI systems from research to deployment, solve complex engineering problems, and deliver data-driven solutions in collaborative environments.

PROFESSIONAL EXPERIENCE

Clemson University, Research Assistant

12/2025 – 08/2026 | Clemson, SC

- Engineered a cohort-based enrollment system supporting **600+ users** (568 students) across monthly cohorts of 30–40, with a free-trial-to-paid conversion funnel and automated entitlement logic.
- Built ML-powered concept quizzes and mastery tracking using **Deep Knowledge Tracing (DKT)** to model student proficiency across SAT concepts.
- Automated monthly student performance reports for teachers using the **GPT API**, converting raw progress data into personalized summaries.
- Built **admin tools** for revenue tracking, discount-coupon management, and micro-influencer campaign performance tied to sales.

Clemson University, Teaching Assistant

08/2026 – Present | Clemson, SC

- TA for CPSC 6600 (Human-Centered AI) and CPSC 8600 (Human-AI Interaction, Collaboration & Teaming), grading assignments and holding office hours for senior/graduate students.
- Reviewed and evaluated student projects in human-AI interaction and collaboration, tracking progress and providing technical feedback via Canvas.
- Contributed to weekly assessment design, helping develop questions for student reports on applied AI/HCI topics.

EDUCATION

M.S. in Computer Science, CLEMSON UNIVERSITY

08/2025 – 05/2027 | Clemson, SC

CGPA: 4.0/4.0

Coursework: Machine Learning, ML-Based Image Synthesis, Data Visualization, Cloud Computing, Reinforcement Learning, Artificial Intelligence

PROJECTS

Access-Atlas

Tech Stack: React, FastAPI, TensorFlow, PyTorch, YOLOv5, OpenStreetMap API

- Developed a full-stack accessibility application for **blind and visually impaired users** using React, FastAPI, and TensorFlow/PyTorch, with deterministic backend data-processing pipelines to deliver **real-time object detection, tagging, and voice-guided navigation**.
- Implemented an object detection pipeline (YOLOv5/SSD) trained on the VizWiz dataset, **achieving 82% detection accuracy** on accessibility-focused labels (ramps, elevators, tactile paths).
- Integrated OpenStreetMap APIs for contextual tagging, **reducing duplicate or noisy tags by 30%** through preprocessing and validation.
- Optimized Python inference and computation paths, **reducing latency by ~25%**.
- Designed **accessibility-first UI** (high contrast, large touch targets, screen reader compatibility), resolving alignment and CORS issues through iterative testing.
- Conducted usability testing with sample users; refinements in alerts and voice guidance **improved navigation confidence scores by 40% vs. baseline**.

Autonomous Driving Trajectory Forecasting (PCD-VLA)

Tech Stack: Python, PyTorch, NVIDIA Alpamayo-1.5-10B, HuggingFace, NumPy, Pandas, SLURM (Palmetto HPC)

- Built a teacher–student trajectory forecasting pipeline on a 200-clip stratified dataset using NVIDIA Alpamayo-1.5-10B as teacher, benchmarking 6 distillation strategies and 3 privileged-learning configurations against a baseline model.
- Ran systematic architecture ablations across distillation, privileged-branch, and navigation-conditioning designs; identified FiLM-based navigation fusion as the highest-impact component, and achieved 3.18m test ADE with the tuned baseline architecture.
- Diagnosed model failure modes through per-class F1 analysis of an 8-class hazard-detection head and confidence-stratified evaluation, using findings to guide model selection and prioritize future development.
- Built a 504-clip robustness benchmark across 7 real-world degradation conditions (occlusion, fog, rain, sensor dropout) to quantify accuracy-robustness trade-offs across model variants.

SKILLS

— **Programming:** Python(NumPy, SciPy, Pandas), Java, JavaScript, SQL

— **Machine Learning:** Deep learning, neural networks, data preprocessing, model evaluation, cross-validation, inference optimization, trajectory forecasting, privileged learning/distillation, synthetic data generation, uncertainty-weighted multi-loss optimization

— **Applied ML / Ed-Tech:** Deep Knowledge Tracing (DKT), student mastery modeling, adaptive learning systems

— **AI/LLM:** GPT-4 API integration, prompt engineering, LLM-powered content generation, RAG (retrieval-augmented generation)

— **Frontend:** Material UI (MUI), design systems, dark-theme/theming architecture, React Hooks, Redux/Context API, responsive UI, accessibility-first design, HTML, CSS

— **Backend & Systems:** REST APIs, authentication (JWT), RBAC, CI/CD (Jenkins), microservices, access-control systems, test automation, DynamoDB, AWS EC2, AWS S3, PM2, Nginx

— **Tools & Platforms:** Git/GitHub, VS Code, PostgreSQL, MongoDB, Docker, simple GUIs, dashboards, user-facing utilities, Cosmos-Drive, HuggingFace